Context selection is covering, not ranking: Minimal faithful representations and amortized partial-order selection

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Abstract

A companion paper shows that context sources for a language model are only partially ordered, and that consequently no single relevance number can rank them faithfully. That is an impossibility result, and it leaves the constructive question open: if one number is not enough, how many are, and can they be computed cheaply? We answer both. For sources modeled as deterministic projections of a latent requirement, we show the Blackwell order is order-isomorphic to the Boolean lattice of the coordinate sets a source reveals, so a source's *coverage set* is a faithful representation. Using the order dimension of that lattice, we prove the minimal faithful vector representation has dimension exactly d, the number of latent coordinates; scalar scoring is the $d = 1$ case, which recovers the companion paper's impossibility as a corollary and sharpens it into an exact requirement. Two structural consequences follow: the non-dominated front is an antichain and can be exponentially large, so a selection budget is unavoidable, and budgeted selection is exactly maximum coverage, whose greedy $(1 - 1/e)$ guarantee is therefore optimal rather than merely conventional. This reframes context selection as a covering problem over latent requirements rather than a ranking problem over documents. We give Probe-Select, which realizes the coverage representation exactly by measurement, and then state the amortization question the reframing makes precise: coverage vectors are predictable from source text, whereas the scalars that existing learned retrievers amortize provably are not. [Planned: a non-circular held-out benchmark against dense, cross-encoder, listwise, and diversity baselines; a learned coverage predictor evaluated by order preservation and end-task regret; and a naturalistic prevalence study.]

1 Introduction

The companion paper [Bilous and Reminnyi, 2026] establishes a negative result: context sources form a Blackwell partial order, incomparable pairs occur and are verifiable on production models, and therefore no query-independent scalar score can rank sources faithfully. It also leaves an explicit gap. Its survey of the design space ends with an empty cell: every method cheap enough for retrieval scale collapses to a total order, and every method that respects source interaction is either symmetric redundancy over embedding geometry or as expensive as direct probing. Nothing is both cheap and faithful to the partial order.

This paper is about that cell. An impossibility result invites exactly one follow-up question, and the companion paper does not ask it: *if one number is not enough, how many are?* We show the question has a sharp answer, that the answer changes what kind of object a selection system should compute, and that the object it identifies is the one thing in this problem that is plausibly learnable.

From ranking to covering. Our starting observation is that for sources modeled as deterministic projections, which is the companion paper's model, the Blackwell order is not an abstract poset. A source reveals a subset of the latent coordinates a task requires, and one source dominates another exactly when its coordinate set contains the other's. The order is therefore order-isomorphic to the Boolean lattice $(2^{[d]}, \subseteq)$, and the natural representation of a source is its *coverage set*. Once that identification is made, the vocabulary of the problem changes. Selection is not sorting documents by a score and cutting; it is choosing a set of sources that *covers* the coordinates the task requires. Context selection is a covering problem, not a ranking problem.

How many numbers are enough. The identification is not merely suggestive; it makes the representation question answerable. Asking for a faithful vector representation is asking for an order-embedding of $2^{[d]}$ into $\mathbb{R}^m$ under the product order, and the least such m is by definition the order dimension of the lattice, which is exactly d [Dushnik and Miller, 1941, Trotter, 1992]. So d numbers suffice, fewer than d provably do not, and the scalar pipeline is the $m = 1$ special case. The companion paper's impossibility theorem falls out as the corollary $m = 1 < d$, now with a quantitative statement of what is missing rather than only that something is.

Why the front needs a budget, and why greedy is optimal. Two consequences follow immediately and neither is visible from the companion paper's framing. First, the non-dominated front is an antichain in the lattice, and by Sperner's theorem [Sperner, 1928] an antichain can have $\binom{d}{\lfloor d/2 \rfloor}$ elements, so "keep everything non-dominated" is not a usable rule at scale and a budget is forced. Second, under a budget the problem is exactly maximum coverage, so the greedy $(1 - 1/e)$ guarantee [Nemhauser et al., 1978] is not a conventional bound imported for respectability: by Feige [1998] no polynomial-time algorithm improves on it unless P = NP. The approximation constant is tight, which is a stronger statement than the companion paper's method sketch was able to make.

Contributions.

1. **Coverage representation** (§3). The Blackwell order over projection sources is order-isomorphic to $(2^{[d]}, \subseteq)$; the coverage set is a faithful representation (Theorem 1).
2. **Exact dimension** (§3.1). The minimal faithful vector representation has dimension exactly d (Theorem 2), with scalar impossibility as a corollary (Corollary 1) and a lower bound on what any learned representation must encode.
3. **Structure of selection** (§3.2). The front is an antichain with a Sperner bound (Proposition 1), and budgeted selection is maximum coverage with a tight $(1 - 1/e)$ constant (Proposition 2).
4. **Exact realization** (§4). Probe-Select computes the coverage order by measurement, with monotone safety and strict generalization of top-k.

5. **The amortization question, made precise** (§5). The dimension bound explains why existing learned retrievers cannot reach the empty cell (they amortize a scalar) and identifies coverage-vector prediction as the target that can.

Relation to the companion paper. We import and do not re-derive: the experiment model, the fixed model rule δ_M, the decision-restricted surrogate $\widehat{\delta}_{\mathcal{D}}$, the finite-sample verification machinery, and the verified anti-monotonicity phenomenon are all established in Bilous and Reminnyi [2026] and used here as given. This paper contributes the representation theory over that model, the structural consequences for selection, and the amortization program. Where the companion paper asks whether scalar selection is broken, this paper asks what to compute instead.

2 Preliminaries

We restate only what the theorems need; see Bilous and Reminnyi [2026] for the full development.

A task T carries a verifier v_T, and $\mathrm{PASS}_M(W, T)$ is the pass rate of a fixed model rule δ_M given context source W. The latent requirement is a vector $S = (S_1, \ldots, S_d)$ of *coordinates*: the private facts a solution must respect. Coordinates are non-degenerate and mutually independent under the prior. A *projection source* W_A, for $A \subseteq [d]$, reveals $(S_i)_{i \in A}$ and nothing else; write $\mathrm{cov}(W_A) = A$ for its coverage set. We write $W \succeq_B W'$ for the Blackwell (garbling) order and take the Blackwell–Sherman–Stein equivalence as given [Blackwell, 1953, Torgersen, 1991].

Verification of empirical dominance and incomparability from finitely many measurements, via exact Clopper–Pearson intervals with a union bound at level η, is the companion paper's machinery and is used unchanged in §4.

3 The coverage representation

Theorem 1 (Coverage isomorphism). *For projection sources, $W_A \succeq_B W_B$ if and only if $B \subseteq A$. Hence $A \mapsto W_A$ is an order isomorphism from $(2^{[d]}, \subseteq)$ onto the Blackwell order restricted to projection sources.*

Proof. If $B \subseteq A$, the map discarding the coordinates in $A \setminus B$ is a deterministic Markov kernel carrying W_A's signal to W_B's, so $W_A \succeq_B W_B$. Conversely, suppose $B \not\subseteq A$ and pick $i \in B \setminus A$. A garbling of W_A is measurable with respect to $\sigma((S_j)_{j \in A})$, which by independence and non-degeneracy of the coordinates is independent of S_i; no such kernel reproduces W_B's signal, which determines S_i. Hence $W_A \not\succeq_B W_B$. Antisymmetry and transitivity transfer from set inclusion. □

The content of Theorem 1 is that the order carries no structure beyond containment of coverage sets. Everything below is a consequence of working in a Boolean lattice rather than in an arbitrary poset.

3.1 How many numbers a faithful representation needs

Definition 1 (Faithful representation). A map φ from sources to $\mathbb{R}^m$ is *faithful* if for all sources W, W',

$$W \succeq_B W' \iff \varphi(W) \geq \varphi(W') \text{ componentwise.}$$

Faithfulness is the property a selection system needs: it may compare representations instead of sources without ever concluding a dominance that does not hold, or missing one that does.

Theorem 2 (Minimal faithful dimension)**.** *The least m admitting a faithful representation of the projection sources on d coordinates is exactly $m = d$. The coverage indicator $\varphi(W_A) = \mathbf{1}_A \in \{0,1\}^d$ attains it.*

Proof. Attainment. $\mathbf{1}_B \leq \mathbf{1}_A$ componentwise iff $B \subseteq A$, which by Theorem 1 holds iff $W_A \succeq_B W_B$; so $\varphi = \mathbf{1}_{(\cdot)}$ is faithful with $m = d$.

Lower bound. A faithful φ is precisely an order-embedding of $(2^{[d]}, \subseteq)$ into $\mathbb{R}^m$ with the product order. By Ore's characterization, the least m for which a finite poset embeds into a product of m chains is its order dimension [Trotter, 1992], and the order dimension of the Boolean lattice $2^{[d]}$ is d [Dushnik and Miller, 1941]. Hence $m \geq d$. □

Corollary 1 (Scalar impossibility, quantified)**.** *A scalar score is faithful if and only if $d \leq 1$. For $d \geq 2$ every scalar mis-orders some pair of sources, recovering the companion paper's incomparability theorem; moreover no representation of dimension $m < d$ repairs it.*

Remark 1 (What this adds to the companion paper). The companion paper proves that *some* faithful scalar fails to exist. Theorem 2 says how far from sufficient a scalar is, and the gap is not a constant: it grows with the number of distinct latent requirements in play. This converts an impossibility into a design specification, namely that a selection system must carry d numbers per source, and it is what makes the amortization question of §5 well posed rather than merely hopeful.

3.2 Consequences for selection

Proposition 1 (The front is an antichain, and can be large)**.** *The set of non-dominated sources in a candidate pool is an antichain in $(2^{[d]}, \subseteq)$. In the worst case its size is $\binom{d}{\lfloor d/2 \rfloor}$, attained by the middle layer [Sperner, 1928].*

Proof. Non-domination is mutual incomparability, which by Theorem 1 is mutual non-containment, the definition of an antichain. Sperner's theorem gives the extremal size. □

So "return the non-dominated set" is well defined but not by itself an algorithm: the front is exponentially large in the worst case. A budget is forced, and with it the question of which members of the front to keep, which the next proposition settles.

Proposition 2 (Budgeted selection is maximum coverage, and greedy is optimal)**.** *Selecting k sources from a front to maximize the number of distinct covered coordinates is an instance of maximum coverage. The greedy rule attains $(1 - 1/e)$ of the optimum [Nemhauser et al., 1978], and no polynomial-time algorithm attains $(1 - 1/e) + \epsilon$ for any $\epsilon > 0$ unless* $\mathrm{P} = \mathrm{NP}$ *[Feige, 1998].*

Remark 2 (Why the constant matters here). A companion-paper draft of this method presented $(1 - 1/e)$ as a standard guarantee and claimed no novelty in it, which understated the situation. Under Theorem 1 the objective *is* coverage, not a submodular surrogate chosen for convenience, and therefore Feige [1998] applies: the greedy step is not merely adequate, it is the best any efficient selector can do. Diversity heuristics defined over embedding geometry [Carbonell and Goldstein, 1998, Ye et al., 2023] optimize a different objective and inherit no such optimality.

4 Probe-Select: realizing the coverage order by measurement

Theorem 2 says what to compute. This section gives the estimator that computes it exactly, at probe cost; §5 asks whether it can be predicted instead.

Verified empirical dominance. Fix δ_M and a probe battery $\mathcal{D}$. Write $W_i \trianglerighteq W_j$ when the companion paper's verification jointly establishes, at confidence $\geq 1-\eta$, that $\widehat{\delta}_{\mathcal{D}}(W_i, W_j) = 0$ up to a tolerance α_{tol} and that $L_{\mathcal{D}}(W_j, W_i) > 0$. Mutually non-dominated sources are verified incomparable and constitute the empirical front. Under Theorem 1, $\trianglerighteq$ is the measured image of coordinate containment, and the probe battery plays the role of the coordinate basis.

Algorithm 1 (Probe-Select). *Input:* candidates $\mathcal{C}$, probe battery $\mathcal{D}$, budget k, model δ_M, level η.

1. Measure PASS(W, T) for all $W \in \mathcal{C}$, $T \in \mathcal{D}$ (n draws per cell); form the Clopper–Pearson intervals once.
2. Build $\trianglerighteq$ from those intervals; no extra model calls.
3. Prune every W verified-dominated by some $W' \in \mathcal{C}$.
4. Let P be the surviving front.
5. If $|P| \leq k$ return P; else greedily add the survivor with the largest verified marginal PASS gain on probe tasks not yet covered, breaking ties by smallest $\widehat{\delta}_{\mathcal{D}}$ to the chosen set, until k are selected.

Proposition 3 (Monotone safety)**.** *Probe-Select never returns a source verified-dominated by an available source. With probability $\geq 1-\eta$ it never discards a source that strictly wins a probe task in favour of one that is nowhere better than it by more than α_{tol}.*

Proof. Step 3 removes only verified-dominated sources and step 5 selects from the survivors. On the joint $(1-\eta)$ event a pruned W has no probe task on which it beats its dominator by more than α_{tol}, and the dominator is retained or Pareto-equivalent to a retained source. □

Proposition 4 (Strict generalization of top-k)**.** *If $\trianglerighteq$ totally orders $\mathcal{C}$, the front is a chain and Probe-Select returns its top k elements, recovering scalar top-k. It therefore extends the scalar pipeline and differs from it only on incomparable pairs, which by Corollary 1 are exactly the pairs a scalar cannot handle.*

4.1 Worked illustration on measured data

Algorithm 1 is implemented (`tokenbench/probe_select.py`, released with the companion paper) and we run it on that paper's measured single-shot record at $n = 40$, adding no new model calls. Take Sonnet-4.6, whose four cells give the candidate PASS profiles

$$W_1 = (100, 100, 0, 100), \quad W_2 = (0, 0, 97.5, 0), \quad W_{1+} = (2.5, 0, 80, 65)$$

on `api_post_ok`, `api_argorder`, `enc_amount`, `trap_store_wire` respectively. No candidate is verified-dominated: W_{1+} beats W_1 on encoding, W_1 beats W_{1+} on the API cells, and W_2 beats both on encoding while losing everywhere else. The front is therefore the whole pool, $\{W_1, W_2, W_{1+}\}$, an antichain of size three. Greedy coverage then takes W_1 (largest verified gain from empty) and W_2 (adds encoding, which W_1 cannot cover), returning $\{W_1, W_2\}$ at $k = 2$ for a routed PASS of 99.4%, against 75.0% for the best fixed source and 36.9% for concatenation.

Table 1 runs this across all six models, and the pattern is uniform in a way worth stating plainly. On every model the front is the entire pool and *nothing is pruned*. This is not a null result but a consequence of the pool: the companion paper constructed W_1 and W_2 to be incomparable, and W_{1+} genuinely wins somewhere, so the candidate set is an antichain by design. What the illustration therefore exercises is the coverage step, not the filter, and the coverage step does the work the filter

Model	|front|	pruned	selected at k=2	routed PASS	concat.
Haiku-4.5	3	none	W_1, W_{1+}	86.2	53.1
Sonnet-4.6	3	none	W_1, W_2	99.4	36.9
Opus-4.8	3	none	W_1, W_2	99.4	59.4
GPT-5.5	3	none	W_1, W_2	100.0	50.0
DeepSeek-V4-Pro	3	none	W_1, W_2	76.9	41.9
Kimi-K2.6	3	none	W_1, W_2	98.8	48.1

Table 1: Algorithm 1 on the companion paper's measured record ($n = 40$ per cell, $\eta = 0.10$, $\alpha_{\mathrm{tol}} = 0.05$). The front is a full antichain on every model, so the pruning step never fires and the coverage step carries the selection. Routed PASS is the per-task maximum over the selected set, i.e. the value under an oracle router; "concat." is the superset W_{1+} alone. Percentages.

cannot: on five of six models it declines to select the superset, whose harm the companion paper verified. The two steps are not redundant, and this pool separates them.

Haiku is the informative exception, and the reason is arithmetic rather than noise: its W_{1+} matches W_2 on encoding (100%) while beating W_1 on the trap cell (92% vs. 90%), so $\{W_1, W_{1+}\}$ genuinely routes to a higher value than $\{W_1, W_2\}$. Anti-monotonicity is a statement about substituting a superset for a component, not about a set that contains both; once W_1 is retained, routing masks the superset's collapse on the API cells. This is a limitation of the routed-PASS endpoint rather than of the selector, and it is why the benchmark of §6 scores assembled context under a realized router rather than an oracle one.

Remark 3 (What this illustration does and does not establish). It establishes that Algorithm 1 runs end to end on real verified measurements and returns the set the theory predicts. It does not establish a benchmark win: the candidate pool has three members and was built to be incomparable, the evaluation cells coincide with the probe battery, and the router is an oracle. Each is addressed in §6, and none of the numbers here should be read as an end-task comparison against retrieval baselines.

Cost. Building $\trianglerighteq$ on $m = |\mathcal{C}|$ candidates costs $m \cdot |\mathcal{D}| \cdot n$ verifier-gated calls (the pairwise comparisons reuse one per-(source, task) PASS table, so cost is linear in m) plus $O(m^2)$ interval comparisons. This is curation scale, not corpus scale, and it is the cost the next section attacks.

5 Amortization: the cheap-and-faithful cell

Why prior amortization cannot reach it. Learned retrievers already amortize "value to the model": REPLUG and REALM train a retriever on the language model's own signal, paying an offline training cost to buy a cheap query-time score [Shi et al., 2023, Guu et al., 2020]. By Corollary 1 this cannot be faithful for $d \geq 2$, and the reason is now precise rather than rhetorical: they amortize an object of dimension 1, and the target has dimension d. Multi-vector late interaction [Khattab and Zaharia, 2020] carries many numbers per document but aggregates to a single query-document score before selection, so the selection-time object is again one-dimensional. The failure is not insufficient training; it is a dimension deficit.

The target this identifies. Theorem 2 names the object to learn: a map $\widehat{\varphi}$ from source text to a coverage vector in $\mathbb{R}^d$, whose induced product order approximates $\trianglerighteq$. This is the amortization the design space is missing, and unlike the scalar case it is not excluded by any result we know.

Training labels are exactly what Algorithm 1 already produces, so the probe tables computed for exact selection are the supervision for the predictor.

What would count as success. Three measurements, in increasing strength: per-coordinate recovery of $\widehat{\varphi}$ against verified coverage; *order preservation*, the fraction of candidate pairs whose dominance or incomparability verdict is preserved by $\widehat{\varphi}$; and *end-task regret*, the PASS gap between selecting with $\widehat{\varphi}$ alone and selecting with full probing. Order preservation is the decisive one, since it is the property Definition 1 requires and the property a scalar provably cannot have.

Remark 4 (Honest statement of the bet). Whether coverage vectors are predictable from source text at useful accuracy is an empirical question and we do not claim it in advance. The theory establishes that the target is the right one and that no lower-dimensional target can work; it does not establish learnability. A negative outcome is informative and reportable: it would locate the obstruction in prediction rather than in representation, which is itself the answer to the companion paper's open problem.

6 Experimental program

[PLANNED. This section is the empirical content of the paper and is not yet run.]

1. **Non-circular selection benchmark (curation scale).** A pool of 8–15 candidate sources with held-out evaluation tasks *disjoint* from the probe battery; endpoint is end-task PASS of the assembled context. Baselines: dense bi-encoder top-k, a cross-encoder reranker, an LLM listwise reranker [Sun et al., 2023], MMR [Carbonell and Goldstein, 1998], and the superset heuristic, all at matched probe budget. Disjointness is the load-bearing design constraint: overlap between probe and evaluation tasks would make the comparison circular.
2. **Coverage-dimension estimate.** Recover the effective d of a candidate pool from the verified order and compare the observed front size against the Sperner bound of Proposition 1.
3. **Learned coverage predictor.** Train $\widehat{\varphi}$ on probe tables; report per-coordinate recovery, order preservation, and end-task regret (§5).
4. **Partial-order re-ranking head at retrieval scale.** Retriever shortlist $\rightarrow$ Probe-Select on the top-m, against reranker-only pipelines on a naturalistic corpus, reporting the *prevalence* of verified incomparability among top-m candidates. This answers the companion paper's open prevalence question.
5. **Sensitivity.** Probe-battery size $|\mathcal{D}|$, per-cell n, and η ablations; the cost–quality frontier.

7 Scope and limitations

The representation results hold in the deterministic-projection model, which is the companion paper's model and the setting in which its evidence was collected. For general stochastic sources the Blackwell order is garbling rather than containment, and no finite-dimensional product-order embedding is faithful in general; Theorems 1 and 2 should be read as sharp statements about projection sources, not about arbitrary experiments. The coordinate count d is a property of a task family together with a candidate pool, not a universal constant, and estimating it is itself an experimental question (item 2 above). Finally, $\trianglerighteq$ is verified against a probe battery, so it inherits the companion paper's decision-restricted reading: conclusions are relative to the fixed model rule δ_M and to the coordinates the battery can resolve.

Reproducibility

The verification machinery, harness, and all measured cells referenced here are released with the companion paper at `https://github.com/abilous-ti/blackwell-llm-context`.

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